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# The coral immune response facilitates protection against microbes during tissue regeneration

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## Abstract

**Increasing physical damage on coral reefs from predation, storms and anthropogenic disturbances highlights the need to understand the impact of injury on the coral immune system. In this study, we examined the regulation of the coral immune response over 10 days following physical trauma artificially inflicted on in situ colonies of the coral *Acropora aspera*, simultaneously with bacterial colonization of the lesions. Corals responded to injury by increasing the expression of immune system-related genes involved in the Toll-like and NOD-like receptor signalling pathways and the lectin–complement system in three phases (<2, 4 and 10 days post-injury). Phenoloxidase activity was also significantly upregulated in two phases (<3 and 10 days post-injury), as were levels of non-fluorescent chromoprotein. In addition, green fluorescent protein expression was upregulated in response to injury from 4 days post-injury, while cyan fluorescent protein expression was reduced. No shifts in the composition of coral-associated bacterial communities were evident following injury based on 16S rRNA gene amplicon pyrosequencing. Bacteria-specific fluorescence in situ hybridization also showed no evidence of bacterial colonization of the wound or regenerating tissues. Coral tissues showed near-complete regeneration of lesions within 10 days. This study demonstrates that corals exhibit immune responses that support rapid recovery following physical injury, maintain coral microbial homeostasis and prevent bacterial infestation that may compromise coral fitness.**

*Keywords*: bacteria, coral, gene expression, immunity, injury

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## Introduction

Disturbances, such as severe tropical storms, sedimentation, predation and algal overgrowth, typically result in wounding and compromised integrity of coral tissues. Other factors, such as entangled fishing line, destructive fishing techniques, anchoring and reef-based tourism activities, also cause significant physical damage to corals (Lamb & Willis 2011; Lamb *et al.* 2014). Ecological surveys of coral reefs have reported that physical injury can be ubiquitous with between 21% and 56% of scleractinian corals injured at any given point in time (Lindsay 2010). Given the recent rise in coral disease (Sokolow 2009) and the likelihood of increasing levels of disturbances that create wounds and enable the entry of microbes into coral tissues, it is important to determine how corals respond to wounding and physical damage, and understand the capacity of corals to recover and regenerate tissues. Although the cellular processes involved in wound healing and tissue regeneration have been well described (summarized below), the dynamics of the underlying molecular responses associated with the co-occurring and vital immune response are less well understood.

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Wound healing is a vital process in maintaining tissue integrity and preventing infection and disease. There are four phases in the healing process in scleractinian corals, similar to the wound healing process in higher organisms. The first step is a wound-sealing phase via coagulation, which results in the formation of a clot in the lesion and is characterized by the release of the granule contents of granular cells, possibly containing transglutaminase (Palmer *et al.* 2011b, 2012). During the second phase, immune cells (eosinophilic granular amoebocytes) that phagocytize microbes and cell debris are recruited to the site of the lesion (Palmer *et al.* 2011b). Additional immune cells and fibroblasts infiltrate and proliferate at the lesion site in the third phase, termed the proliferation phase, resulting in the formation and reorganization of an epidermal-like layer (Palmer *et al.* 2011b). Phase four encompasses wound maturation, which can occur within 48 h (Palmer *et al.* 2011b), and is generally followed by remodelling of the epidermal cells into a new epithelium. Despite an understanding of the short-term cellular mechanisms of wound healing (Palmer *et al.* 2011b) and lesion regeneration (Bak & Steward-van Es 1980; Bak 1983; van Woesik 1998; D'Angelo *et al.* 2012; Denis *et al.* 2013), the longer-term response and underlying molecular immune mechanisms in corals remain to be elucidated.

Corals possess a suite of innate immune defence mechanisms that enable them to maintain a close symbiosis with associated bacterial communities and eliminate potentially harmful microbes. Low, but consistent background levels of antimicrobial peptide (AMP) expression result in the maintenance of a healthy microbial community in several invertebrates, including cnidarians (Fraune *et al.* 2011; Kounatidis & Ligoxygakis 2012), and recently, an AMP was isolated from the scleractinian coral *Pocillopora damicornis* (Vidal-Dupiol *et al.* 2011). AMPs exhibit bactericidal properties by forming pores in microbial cell walls or by binding to intracellular molecules essential for a microbe's survival (Brogden 2005). During infection, AMP expression is generally upregulated upon the detection of microbe-associated molecular patterns (MAMPs) by pattern recognition receptors (PRRs), such as Toll-like receptors (TLRs). TLRs and nucleotide-binding oligomerization domain (NOD)-like receptors (NLRs) signal via the JNK, MAPK p38 and NF-κB pathways, resulting in the transcription of target genes involved in immunity, cell survival and proliferation. A repertoire of genes encoding TLRs and NLRs, as well as components of their downstream signalling pathways, has recently been described in corals (Miller *et al.* 2007; Shinzato *et al.* 2011), although no studies have examined the expression of these genes following injury.

Invertebrates also employ several rapidly inducible immune mechanisms to combat infections, such as transglutaminase. In arthropods, this enzyme polymerizes haemolymph proteins into large aggregates, resulting in coagulation at the site of injury (Cerenius *et al.* 2010). Recently, evidence of transglutaminase activity in corals was reported (Palmer *et al.* 2012), which, when combined with the identification of hemolectin homologues in *A. millepora* (Grasso *et al.* 2008, 2011), suggests the presence of a coagulation system in corals. Another rapidly inducible immune mechanism is the highly conserved melanization cascade, or proPO system, which has recently been documented in both gorgonian and scleractinian corals (Mydlarz *et al.* 2008; Palmer *et al.* 2008). In this system, the inactive pro-enzyme pro-phenoloxidase (proPO) is cleaved into the active phenoloxidase (PO) following the detection of MAMPs (Mydlarz *et al.* 2006; Cerenius *et al.* 2010). PO activity results in the deposition of melanin, which immobilizes microbes (Cerenius *et al.* 2010), while at the same time highly reactive and toxic quinone intermediates that attack the invading organisms are produced (Cerenius *et al.* 2010). PO is also involved in hardening of the clot formed during the coagulation of the haemolymph following wounding (Cerenius *et al.* 2010; Kounatidis & Ligoxygakis 2012). Melanin deposits have recently been found around the lesions of injured corals, confirming a role for PO in the coral wound healing process (Palmer *et al.* 2011b). In many invertebrates, components of the complement system, such as complement C3, factor B (Bf), lectins and mannose-binding lectin (MBL)-associated serine protease (MASP), have been discovered (Mydlarz *et al.* 2006; Cerenius *et al.* 2010). In this system, MBLs have been proposed to bind to the microbial cell wall and activated MASPs cleave C3 and Bf. In association, these two proteins form a C3 convertase that efficiently cleaves C3 and deposits C3b on the microbe's surface. The opsonin properties of C3b ensure enhanced phagocytosis and elimination of the microbe by specialized immune cells. In addition, spontaneous hydrolysis of C3 and the subsequent binding of C3b to a microbial cell wall could lead to the formation of a membrane attack complex (MAC), which forms pores in microbial cell walls resulting in cell lysis. In corals, the MBL millectin was found to cause the agglutination of bacteria in vitro and expression was increased upon challenges with immunostimulants (Kvennefors *et al.* 2008). C3 has also been shown to exhibit antibacterial properties as it is upregulated in response to bacteria (Brown *et al.* 2013) as well as following trauma and injury (Kvennefors *et al.* 2010). Millectin is also associated with *Symbiodinium*, suggesting that it may play a role in symbiosis (Kvennefors *et al.* 2008, 2010). In addition, genes encoding proteins containing a MAC/PF (membrane

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attack complex/perforin) domain have been identified in Acr*opora millepora* (Miller *et al.* 2007).

Evidence of the presence and activation of several innate immune system pathways in corals suggests they possess complex wound healing and immune systems, encompassing coagulation, phagocytosis and antimicrobial properties, as seen in higher organisms. However, little is known about the genes involved in the immune response that accompanies cellular wound healing and tissue regeneration. In this study, we investigated the process of wound healing in a scleractinian coral species over a 10-day period to determine the response of the innate immune system to tissue damage and its role in tissue repair. Specifically, we assessed the expression dynamics of immune system-related genes and proteins as well as bacterial colonization patterns and coral-associated bacterial community structures, during the wound healing and tissue regeneration process. Our results show that tissue damage elicits a dynamic response in corals, involving various components of the coral immune system and tissue repair process, sufficient to prevent bacterial infection.

## Materials and methods

### *Field experimental design and sample collection*

Two large patches (approximately 2 m in diameter) of the scleractinian coral *Acropora aspera* were located on the reef flat of Heron Island, Great Barrier Reef, Australia (23°26′31.20″S, 151°54′50.40″E), and tagged on 6 February 2011. Each patch consisted of a collection of fragments without connecting tissues, with the dead bases of each fragment covered in algae and CCA (Fig. S1, Supporting information). As branches within a patch appeared to have been connected in the past, we considered each patch to be a unique genotype. Tissue was removed from 14 branches from each of the three sides of each selected patch ($n = 42$ damaged branches per genotype) using a high-pressure airgun. The artificially inflicted injuries simulated physical damage resembling relatively clean lesions consistent with hydrodynamic or human-caused damage, as well as certain feeding scars (e.g. corallivorous butterfly fish and crown-of-thorn starfish). Lesions were an approximately 1-cm-wide strip along three-fourth of the branch length (~7.5 cm). Samples were taken 0.7 (16 h), 1, 2, 3, 4, 7 and 10 days post-injury at 16.00 h each sampling day, with the exception of the first time point at 08.00 h. Care was taken that all samples were sourced from different fragments and can therefore be considered independent units. At each sampling time, we collected 12 independent nubbins per genotype: two injured and two undamaged and healthy-looking nubbins from independent fragments on each of the three sides ($n = 6$ per health state) of each patch. Of these, one injured and one healthy-looking nubbin from each side were snap-frozen in liquid nitrogen, and the remaining nubbins were fixed in 4% paraformaldehyde in phosphate-buffered saline (PBS) solution (Table S1, Supporting information for an overview of samples used for each analysis).

### *Fluorescence in situ hybridization*

Fixed coral nubbins were decalcified in 20% (w/v) Na-EDTA (pH 8.0) and subsequently cut into 3–5 pieces of equal length prior to histological processing. Samples were processed sequentially in 70%, 80%, 90%, 95% and 100% (two washes) ethanol, followed by three washes in xylene and three washes in Paraplast wax, each for 60 min, overnight in a Shandon Histocentre three automated tissue processor, prior to embedding in Paraplast wax. Paraffin blocks were sectioned at 7 μm and collected on Superfrost Plus slides (Menzel, Germany). Universal bacterial probe EUB338 (Amann *et al.* 1990), coupled to the fluorescent label ATTO647 (Thermo Fisher Scientific, USA), was used in a standardized FISH protocol (Ainsworth *et al.* 2006). Tissue sections were dewaxed in xylene (3 × 10 min), dehydrated in absolute ethanol (3 × 3 min) and air-dried. Sections were covered with hybridization buffer (900 mM NaCl, 20 mM Tris–HCl, 35% (v/v) formamide and 0.01% SDS in ultrapure water), and probes were added to a final concentration of 25 ng/μL, incubated at 46 °C for 2.5 h and washed in 50 mL wash buffer (70 mM NaCl, 20 mM Tris–HCl, 1 mM EDTA and 0.01% SDS in ultrapure water) at 48 °C for 15 min. Sections were thoroughly rinsed with ice-cold ultrapure water to remove excess salts, air-dried and embedded in Citifluor AF1 (Citifluor, UK). An LSM710 confocal laser scanning microscope (Carl Zeiss, Germany) combined with spectral emissions profiling was used to visualize tissue-associated FISH-labelled bacterial communities as per Ainsworth *et al.* (2006).

### *Messenger RNA isolation*

Frozen nubbins of *A. aspera* were crushed in a pre-chilled (with liquid nitrogen) stainless steel mortar and pestle using a hydraulic press. Messenger RNA (mRNA) was isolated from the crushed coral using the Dynabeads mRNA DIRECT kit (Invitrogen Dynal, Norway) according to a modified protocol based on the manufacturer's recommendations. Approximately 100 mg of crushed coral was added to 400 μL lysis buffer, incubated on a vortex at low speed for 7 min and centrifuged for 2 min at 12 000 *g*. Supernatant was

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added to oligo(dT)-Dynabeads (prewashed in lysis buffer) and incubated on the vortex at medium speed for 8 min to allow mRNA annealing. Tubes were placed on a DynaMag-2 magnetic particle concentrator for 5 min, and supernatant was removed. Using the DynaMag-2, oligo(dT) Dynabead/mRNA complexes were washed twice with 300 μL of Buffer A and subsequently twice with 400 μL of Buffer B. Complexes were resuspended in 27 μL ice-cold 10 mM Tris–HCl, incubated at 80 °C for 2 min and rapidly cooled down on ice. Oligo(dT)-Dynabeads were concentrated on the DynaMag-2, and mRNA-containing supernatant was collected and stored at −80 °C.

### *Gene expression analysis*

Expression levels of 20 immune system-related genes (Siboni *et al.* 2012) and four reference genes (Seneca *et al.* 2010) were analysed using the GenomeLab GeXP Start Kit and the CEQ-8800 Genetic Analysis System (Beckman-Coulter, USA) following the protocol described by Siboni *et al.* (2012). cDNA was generated from 9 ng of mRNA, and forward primer concentrations were 200 nM. Reverse primer concentrations were optimized for the multiplex to ensure signals in the electropherogram were within the CEQ-8800 detection range: 500 nM for *TRAM*, *TIR-1*, *MEKK1*, *TRAF6*, *Tx60*, *ERK2*, *Bf*, *MAPK p38*, *Millectin*, *ctg_1913*, *apextrin*, *cJun*, *HL-1*, *HL-2*, *HL-3*, *CTL-1* and *CTL-2*; 250 nM for *GAPDH* and *RPL9*; 62.5 nM for *NF-κB* and *C3*; 12.5 nM for *cFos* and *ATF4/5*; and 3.75 nM for *RPS7* (Table S2, Supporting information). PCR products were prediluted 1:20 prior to loading on the CEQ-8800. Data were filtered and analysed using the GeXP Fragment Analysis module and eXpress Profiler software (Beckman-Coulter, USA) (Siboni *et al.* 2012). Gene expression levels were normalized to an internal control ($Kan^R$) and to the geometric mean of the expression levels of three housekeeping genes (*RPL9*, *GAPDH* and *ctg_1913*) selected using geNorm (Vandesompele *et al.* 2002). Results were obtained for three technical replicates per sample.

### *Clarification of terminology*

In a number of previous studies, the term phenoloxidase (PO) activity has been used to describe levels of the active enzyme detectable in a sample and the term prophenoloxidase (proPO or PPO) activity has been used to describe the sum of active and potential enzyme activity. The latter activity is revealed by converting the inactive pro-phenoloxidase zymogen to the active PO form through trypsin-mediated experimental cleavage. As proPO is a pro-enzyme or zymogen and thus lacks any form of activity, here we use the term tpPO (total potential phenoloxidase) activity, rather than proPO activity, to more accurately reflect the parameter quantified by the assay. In our study, both PO and tpPO activity refer to tyrosinase-type phenoloxidase activity that is assayed using dopamine as substrate.

### *Protein and enzyme assays*

Approximately 120 mg of each frozen crushed coral sample was added to 500 μL phosphate-buffered saline (50 mM, pH7.8) (Sigma-Aldrich, Australia) and lysed using Lysing Matrix C and a FastPrep-24 (MP Biomedicals, USA) at 4.0 MΩ/s for 1 min. Samples were centrifuged for 45 min at 12 000 *g* at 4 °C, and supernatants were collected and stored at −80 °C. Total protein content was determined for each sample using the DC Protein Assay (Bio-Rad, USA) following the manufacturer's instructions. Absorbances were read at 750 nm using a SpectraMax M2 (Molecular Devices, USA).

The assay to measure total potential phenoloxidase (tpPO) activity followed those outlined in Palmer *et al.* (2011a) with some modifications. Twenty microlitres of coral tissue lysate was loaded in triplicate into wells of a 96-well plate, and phosphate-buffered saline (50 mM, pH7.8; 40 μL) and 25 μL 0.1 mg/mL trypsin were added. For the analysis of phenoloxidase (PO) activity, trypsin solution was substituted by 25 μL double-distilled water. Reaction mixtures were incubated for 20 min to allow for the activation of pro-phenoloxidase by trypsin, and 30 μL of 10 mM dopamine hydrochloride (Sigma-Aldrich, Australia) was added. As a control, 20 μL of PBS was used. After a 15-min incubation, absorbance was measured at 490 nm every 5 min for a 45-min period using a SpectraMax M2. Data were independently obtained in triplicate. All data were normalized to total protein content.

For chromoprotein (CP) and fluorescent protein (FP) concentrations, 20 μL of tissue lysate sample was added to each well of a black, clear bottom 384-well plate in triplicate. Expression of chromoprotein was analysed by measuring the absorbance at 588 nm using a SpectraMax M2 (Alieva *et al.* 2008). Fluorescence spectra were analysed by measuring the emission wavelengths between 400 and 700 nm, with a 5-nm resolution, upon excitation of fluorescent proteins at 280 nm using a SpectraMax M2. Data were obtained in triplicate. Spectra were analysed for relative contributions of individual FPs (cyan (CFP), green (GFP) and red (RFP)) to total fluorescent protein concentration using a multiple regression analysis. As standards of the three FP colour classes, emission characteristics of the purified *A. millepora* FPs expressed in bacteria (Alieva *et al.* 2008) were used. An additional component in the multiple regres-

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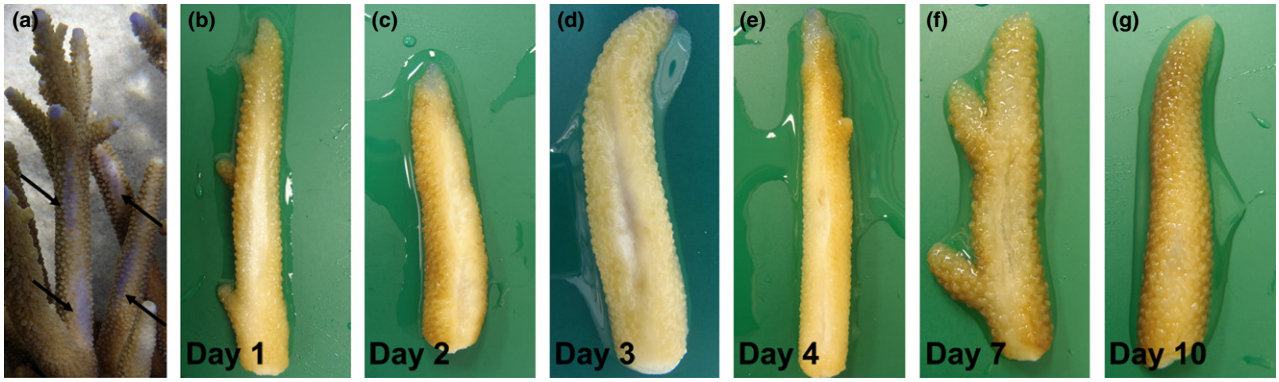


**Fig. 1** Macroscopic analysis of injured coral. (a) Pigmentation response (arrows) at lesion site 7 days post-injury, (b–g) Wound healing in *Acropora aspera* during the 10-day period following injury. Photographs are representative images of (a) injured corals in situ and (b–g) decalcified nubbins of injured corals.

sion was the exponentially decaying background, which was modelled using the formula $B = F_{450}/\exp(k*(w-450))$. In this formula, $F_{450}$ is the reading at 450 nm (outside of the emission spectra of all the three FPs), w is the wavelength in nanometres, and k is the scaling coefficient, which was set to 0.1 throughout the analyses as it resulted in best fit of the models. Integrated background-subtracted fluorescence attributable to each FP component was then multiplied by a coefficient reflecting the relative excitation at 280 nm (set to 1 for CFP, GFP and RFP) to obtain relative proportions of fluorescent proteins. These proportions were then multiplied by the background-subtracted integrated total fluorescence and standardized to total protein concentration, to obtain values that are directly proportional to the concentration of each FP in the sample.

### *Statistical analysis*

Differences in the dependent variable's gene expression, protein concentration and PO activity data between healthy and injured samples were analysed for differences using general linear models (GLM) with health state and time as categorical predictors. Genotype was not included as a categorical predictor in the model as no genotype effects were found, with the exception of *apextrin* and *cFos*. The Fisher LSD post hoc test was used to determine the statistical significance of differences between treatments at different time points. Differences were considered significant when $P < 0.05$. All analyses were conducted using the STATISTICA 12 statistical software package.

### *16S rRNA gene amplicon pyrosequencing*

Samples of approximately 100 mg of crushed frozen coral were aseptically transferred to 1.5-mL microcentrifuge tubes, and total genomic DNA was extracted using the PowerPlant DNA Isolation kit (MO BIO Laboratories, USA) following the manufacturer's instructions. Extracted DNA was quantified using a NanoDrop (ThermoScientific) and shipped to the Research and Testing Laboratory (RTL; Texas, USA). Bacterial 16S rDNA was amplified using the 28F (5′-GAGTTTGAT CNTGGCTCAG-3′) and 519R primer (5′-GTNTTACN GCGGCKGCTG-3′) sets, followed by sequencing on a Roche GS FLX+ 454 pyrosequencer at RTL. Sequence fasta and quality files were obtained from RTL and processed using the QIIME pipeline (Caporaso *et al.* 2010). The split_libraries.py script was used to remove poor quality (<25) sequences, reads of <200 bp or >550 bp in length, primers and barcodes and to add a sample identification to each sequence. Using UCHIME (Edgar *et al.* 2011), chimeric sequences were identified against the chimera-free 16S rRNA gene sequence database (SILVA Gold Alignment Database) and removed. Using the chimera-free fasta file, numbers of reads per sample were calculated and samples were normalized to 600 reads per sample. Sequences of 97% similarity were clustered using uclust (Edgar 2010), with each cluster representing an operational taxonomic unit (OTU). The most abundant sequence per cluster was selected as representative sequence, and Greengenes taxonomy was assigned using the Greengenes database (version gg_13_5) and BLAST (75% similarity). QIIME pipeline was used to generate OTU tables, and tables were analysed for relative abundance. Alpha diversity metrics (total observed species (OTU), predicted species (chao1), Shannon–Weiner diversity and Fisher's alpha) were generated from OTU tables. Beta diversity was analysed using a two-dimensional principal coordinate analysis, based on UniFrac distances generated from weighted and unweighted UniFrac analysis using PRIMER 6 (PRIMER-E, UK). PERMANOVAs on square root-trans-

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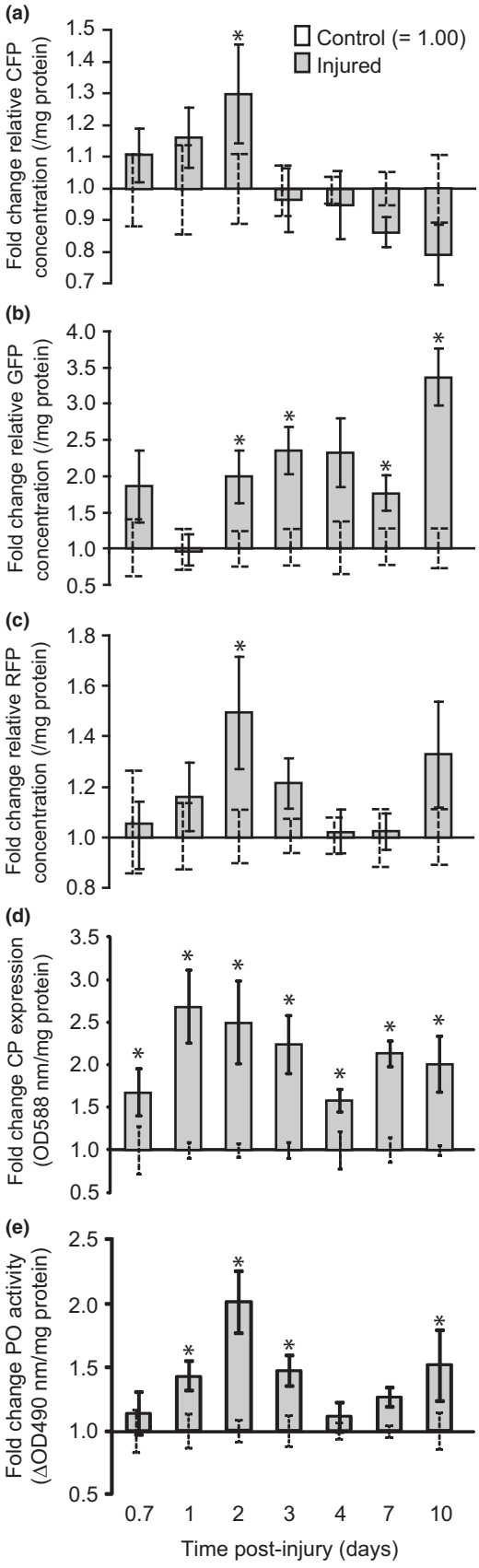


**Fig. 2** Temporal GFP-like protein expression and phenoloxidase activity in response to injury. (a–e) Temporal expression patterns of (a) cyan fluorescent protein (CFP), (b) green fluorescent protein (GFP), (c) red fluorescent protein (RFP), (d) chromoprotein (CP) and (e) phenoloxidase (PO) activity. Data are relative to control. Asterisk (*) indicates statistical significance of $P < 0.05$. Dashed error bars indicate standard error for control, and solid error bars indicate standard error for injured coral samples.

formed OTU tables as well as weighted and unweighted UniFrac distances were performed to test for significant differences between coral-associated bacterial communities using PRIMER 6 (9999 permutations). The complete data set was deposited in the NCBI Sequence Read Archive (SRA) database under study Accession no. SRP052969.

## Results

### *Macroscopic and cellular assessment of coral recovery*

Following artificially inflicted physical damage, a pigmentation response was observed within 16 h (0.7 days) at all lesion sites in *A. aspera* and was visible until the end of the monitoring period, 10 days post-injury (Fig. 1a). In the majority of injured coral fragments (11 of 12), almost complete tissue regeneration occurred within 10 days, showing near-normal macroscopic tissue morphology and pigmentation (Fig. 1b–g). No identifiable signs of macroscopic microbial colonization or overgrowth of tissues were associated with the lesion.

### *Molecular responses following physical damage*

Expression of fluorescent proteins from all three classes analysed (CFP: $P = 0.04$; GFP: $P < 0.01$; RFP: $P < 0.01$) was significantly upregulated 2 days post-injury (Fig. 2a–c). The concentration of green fluorescent protein (GFP) was significantly upregulated (by up to 3.4-fold) on all subsequent days (Day 3: $P = 0.02$; Day 7: $P = 0.03$; Day 10: $P < 0.01$), except Day 4 ($P = 0.09$) (Fig. 2b). CFP and RFP were not differentially expressed in wounded tissues compared to controls on all other time points analysed (Fig. 2a,c). Concentrations of GFP and RFP were relatively low, however, and results should be interpreted with caution. The visual presence of blue–purple pigmented tissue in the coral lesion areas was suggestive of the upregulation of a blue chromoprotein (CP) (Dove *et al.* 2001), and accordingly, concentrations of this protein were found to be upregulated within hours (0.7 days) following injury ($P = 0.04$) and remained upregulated compared to control coral nubbins throughout the 10-day study period ($P < 0.01$ at each time point) (Fig. 2d). Temporal pat-

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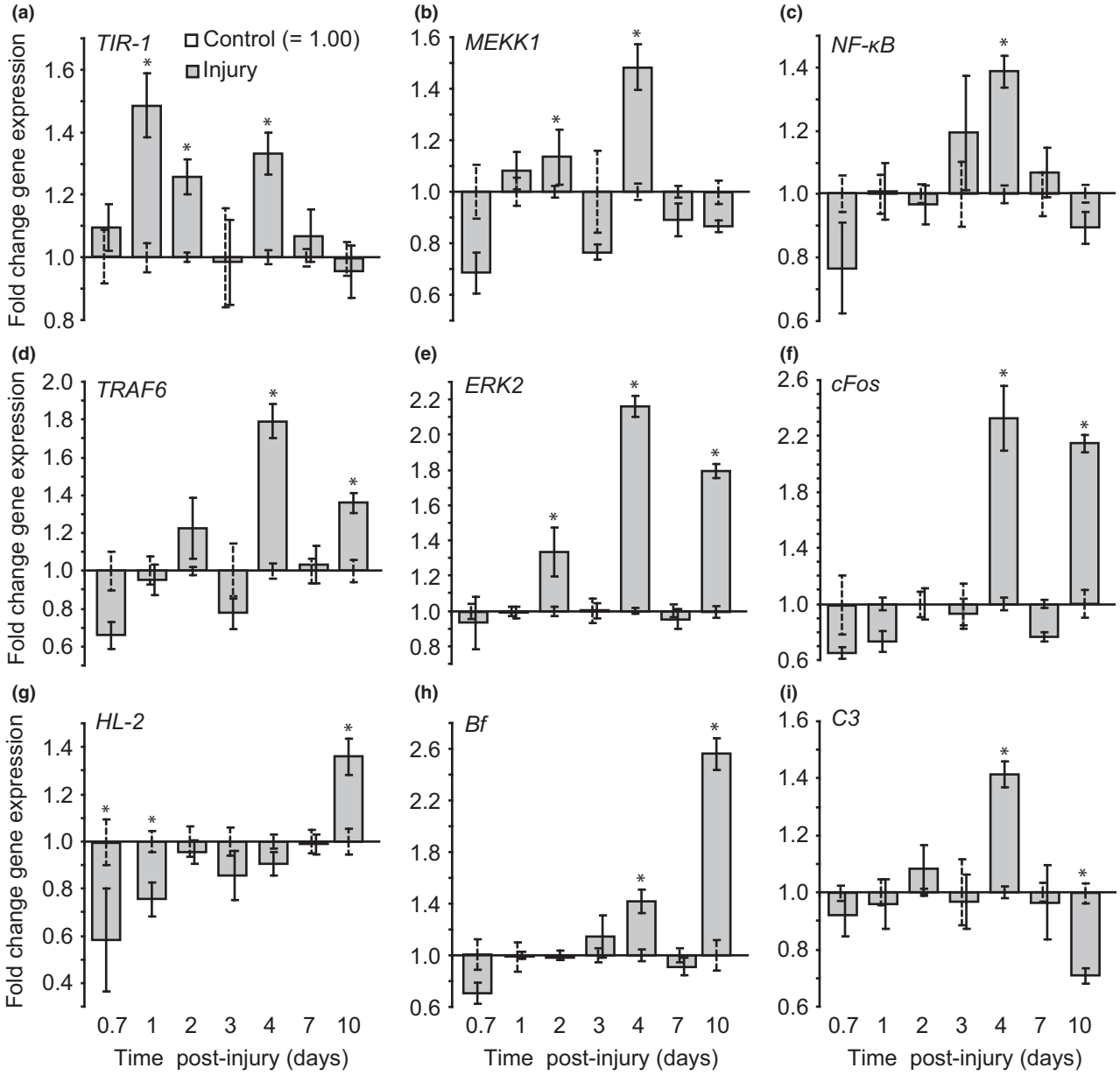


**Fig. 3** Temporal expression patterns of immune system-related genes in response to injury. (a–i) Temporal expression patterns of genes involved in PRR signalling (a) Toll-like receptor homologue *TIR-1*, (b) *MEKK1*, (c) *ERK2*, (d) *NF-κB*, (e) *TRAF6*, (f) *cFos* and immune effectors (g) *HL-2*, (h) *Bf* and I) *C3*. Data are relative to control. Asterisk (*) indicates statistical significance of $P < 0.05$. Dashed error bars indicate standard error for control, and solid error bars indicate standard error for injured coral samples.

terns in CP tissue concentration showed two peaks: the first within 1 day post-injury and a second lower peak on day 7 (Fig. 2d). Phenoloxidase (PO) activity increased significantly between days 1 and 3 post-injury ($P < 0.01$), with peak activity evident 2 days post-injury ($P < 0.01$), compared to control samples (Fig. 2e). Subsequently, activity decreased in injured corals, reaching levels similar to those of controls at days 4 and 7 ($P = 0.54$), although a second peak in PO activity was evident at 10 days post-injury ($P < 0.05$) (Fig. 2e). Total potential phenoloxidase activity mirrored this temporal pattern ($r = 0.99$) (Fig. S2a, Supporting information).

At 0.7 days (16 h) post-injury, expression patterns of a suite of immune system-related genes demonstrated either non-significant trends for reduced gene expression or significant downregulation (*ATF4/5:* $P = 0.04$; *millectin:* $P < 0.01$) for all genes of interest, except *TIR-1* (Fig. 3, Fig. S2b,e, Supporting information). The putative Toll-like receptor homologue *TIR-1* was the first gene observed to be significantly upregulated (1.49-

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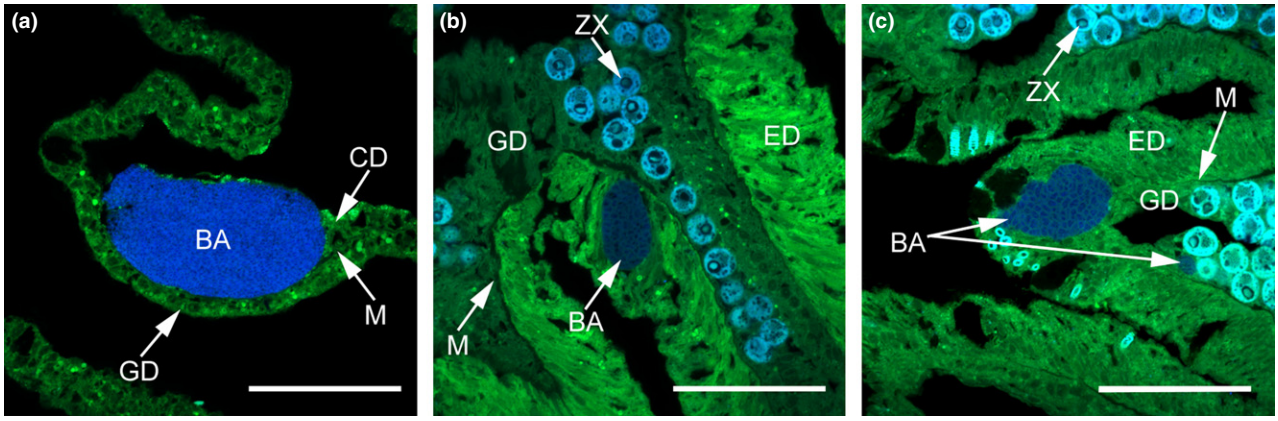


**Fig. 4** Microscopic analysis of coral-associated bacteria in injured coral. (a–c) Localization of bacterial aggregates in coral tissues following injury: aggregates within (a) calicodermal and gastrodermal cell layers, (b) ectodermis and (c) both ectodermal and gastrodermal cell layers. Bacteria were visualized using the bacteria-specific probe EUB338. Magnification: 63×. Scale bar: 50 μm. BA, bacterial aggregate; ZX, zooxanthellae; E, ectodermis; G, gastrodermis; M, mesoglea.

**Table 1** Overview of alpha diversity metrics of coral-associated bacterial communities

| | Total sequences | Rarefied sequences | OTU | chao1 | Shannon index | Fisher's alpha |
|---|---|---|---|---|---|---|
| Day 2 | | | | | | |
| Healthy | 22 761 | 4282 | 112 | 114 | 5.36 | 21.05 |
| Injured | 17 477 | 4282 | 149 | 149.52 | 5.98 | 29.99 |
| Day 3 | | | | | | |
| Healthy | 15 565 | 4282 | 60 | 66.43 | 4.55 | 9.88 |
| Injured | 15 516 | 4282 | 153 | 170.88 | 4.90 | 31.00 |
| Day 4 | | | | | | |
| Healthy | 7261 | 4282 | 79 | 99.00 | 4.81 | 13.75 |
| Injured | 10 469 | 4282 | 87 | 102.00 | 4.72 | 15.45 |
| Day 7 | | | | | | |
| Healthy | 7458 | 4282 | 77 | 84.00 | 4.55 | 13.33 |
| Injured | 13 398 | 4282 | 90 | 1054.25 | 4.87 | 16.11 |
| Day 10 | | | | | | |
| Healthy | 4282 | 4282 | 197 | 208.5 | 6.19 | 42.65 |
| Injured | 6769 | 4282 | 101 | 146.15 | 3.66 | 18.54 |

fold) within 1 day post-injury ($P < 0.01$) and remained upregulated until Day 2 ($P = 0.03$). Following a return to control levels 3 days post-injury (Fig. 3a), a second upregulation in *TIR-1* expression (1.33-fold) was observed on Day 4 ($P < 0.01$), but expression levels did not differ from those of controls by 7 days post-injury (Fig. 3a). Genes involved in signalling downstream of TLRs also showed variable temporal expression profiles, with upregulation in expression on Day 2 (*MEKK1:* 1.13-fold, $P = 0.03$; *ERK2:* 1.33-fold, $P < 0.01$) and Day 4 (*NF-κB:* 1.39-fold, $P < 0.01$; *TRAF6:* 1.79-fold, $P < 0.01$; *cFos:* 2.33-fold, $P = 0.02$; *MEKK1:* 1.48-fold, $P < 0.01$; and *ERK2:* 2.16-fold, $P < 0.01$) (Fig. 3b–f). Some of these genes were also upregulated on Day 10 (*TRAF6*: 1.36-fold, $P = 0.04$; *ERK2*: 1.80-fold, $P < 0.01$; and *cFos*: 2.14-fold, $P < 0.01$), whereas this was not observed for other genes involved in TLR signalling (Fig 3b–f). No changes in the expression of *cJun* were observed, but *ATF4/5* (Day 0.7: $P = 0.04$; Day 3: $P = 0.02$; Day 10: $P = 0.01$) and *MAPK p38* (Day 4: $P = 0.04$; Day 10: $P = 0.01$) were downregulated (up to 0.79- and 0.76-fold, respectively) (Fig. S2b–d, Supporting information).

*Millectin* was consistently downregulated in injured corals (up to 0.27-fold on Day 0.7, and at Day 1, 3, 4 and 7: $P < 0.01$) throughout the 10-day sampling period (Fig. S2e, Supporting information). Downregulation was also observed for the homologue of hemolectin *HL-2* at early time points (Day 0.7: $P < 0.01$; Day 1: $P < 0.05$); however, this gene was significantly upregulated on day 10 (1.36-fold, $P < 0.05$) (Fig. 3g). Expression of the C-type lectin *CTL-2* was upregulated in response to physical damage compared to control corals, which had

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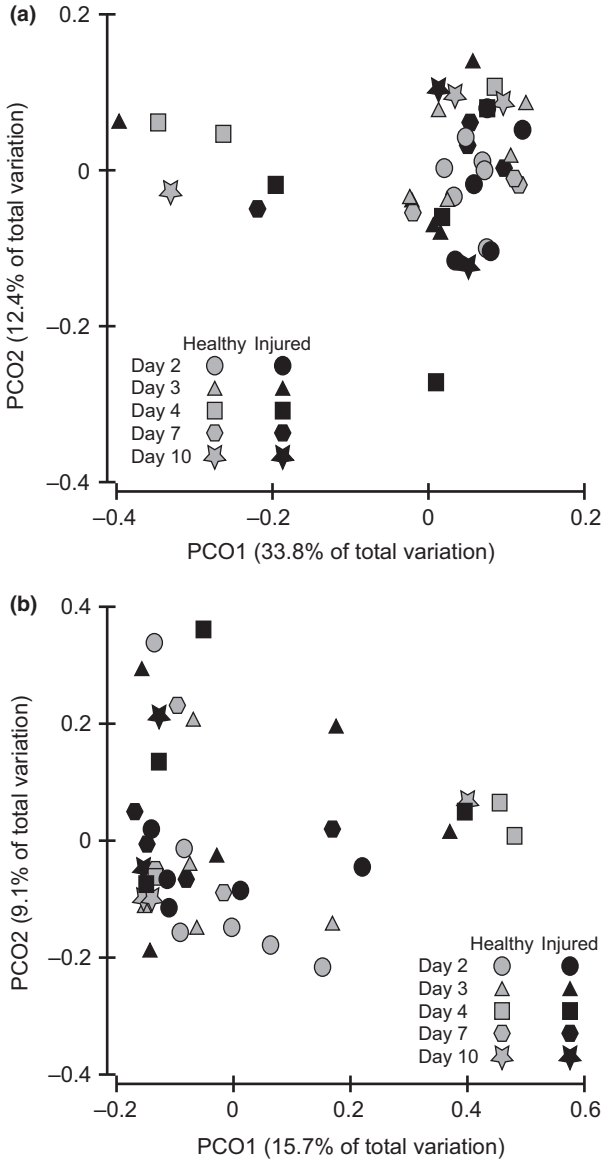


**Fig. 5** Principal coordinate analysis representing bacterial community profiles of healthy and injured corals based on 16S rRNA amplicon sequencing. Analyses were based on (a) weighted and (b) unweighted UniFrac distances.

*CTL-2* levels that were either very low or were undetectable (Fig. S2h, Supporting information). The expression of *Bf* and complement *C3* were found to follow similar patterns to several of the TLR downstream signalling molecules, being significantly upregulated at Day 4 (*Bf:* 1.42-fold, $P < 0.05$; *C3:* 1.41-fold, $P < 0.01$) (Fig. 3h–i). Ten days post-injury, expression levels of *Bf* were also significantly increased (2.56-fold, $P < 0.01$), which was in contrast to a downregulation in expression of *C3* (0.71-fold, $P < 0.01$) (Fig. 3h–i). Genes that are potentially involved in the formation of the membrane attack complex were expressed at either similar levels to the control samples or downregulated (*apextrin* downregulated by up to 0.49-fold on days 1 and 2 ($P < 0.05$) and days 4 and 7 ($P < 0.01$); *Tx60* downregulated by 0.34-fold on Day 3 ($P < 0.01$); Fig. S2f,g, Supporting information). Full details of statistical outcomes are presented in Table S3 (Supporting information) (main effects) and Table S4 (Supporting information) (Fisher's LSD post hoc test).

### *Bacterial community response following physical damage*

Examination of the tissues using FISH with general bacteria-specific probes did not show signs of microbial invasion of tissues or exposed skeletal spaces following artificially induced injury. Aggregates of bacteria were observed within the calicodermal and gastrodermal cell layers that comprise the aboral tissues and within ectodermal and gastrodermal cell layers that comprise oral tissues of both healthy and injured corals during FISH examination (Fig. 4a–c). There were no observable differences in number or size of bacterial aggregation in the tissues from injured vs. healthy corals.

Amplicon-based 16S rRNA gene amplicon pyrosequencing was used to profile the bacterial community of both healthy and injured corals. Overall, injury had no significant effect on species richness (Chao1) or alpha diversity (Shannon index, Fisher's alpha, number of unique OTUs; Table 1). Similarly, principal coordinate analysis of the beta diversity of bacterial communities did not reveal any changes as a result of injury or over time (Fig. 5a,b). Results of a PERMANOVA confirmed that there were no changes in the coral-associated bacterial communities due to tissue damage ($P = 0.26$), sampling time ($P = 0.47$) or damage over time ($P = 0.34$). In addition, no changes were observed in abundance of the species present (damage: $P = 0.10$; sampling time: $P = 0.36$; interaction: $P = 0.19$), nor did we observe a change in bacterial species associated with the corals (damage: $P = 0.24$; sampling time: $P = 0.31$; interaction: $P = 0.36$).

## Discussion

Investigations into wound healing in invertebrates have primarily focused on cellular and molecular wound repair and tissue regeneration processes, with limited attention to the immune responses induced by wounding. Here, we characterize the coral immune response to tissue damage and concurrent effects on coral-associated bacterial communities in the scleractinian coral *Acropora aspera* over a 10-day period.

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### *Tissue regeneration following physical damage*

Near-complete wound healing and restored *Symbiodinium*-related pigmentation was observed after 10 days following artificially induced injuries in *A. aspera*, which contrasts with the slower regeneration times of 60 days to up to 19 months previously reported for other scleractinian corals (Bak & Steward-van Es 1980; Bak 1983; Meesters & Bak 1994; van Woesik 1998; Denis *et al.* 2013; Horwitz & Fine 2014). Many intrinsic and extrinsic factors, for example species, colony morphology, genotype, wound characteristics and environmental factors, are known to affect the regeneration capacity of corals (Henry & Hart 2005; Denis *et al.* 2013), providing possible explanations for the much more rapid regeneration observed in this study.

### *The coral immune gene response to physical damage*

Our study of the coral immune response to physical damage at the transcriptome level highlights the longer time frame of the response than is typically characterized in studies of other invertebrates, which have tended to focus on the initial 2 days post-injury. These studies report that many genes are upregulated in the first 2 days, including those involved in the stress response, metabolism and cell adhesion (Altincicek & Vilcinskas 2008), as well as in the immune system, such as antimicrobial peptides (AMP) (Erler *et al.* 2011), membrane attack complex (MAC) proteins and a NADPH oxidase (Altincicek & Vilcinskas 2008). In contrast, the transcriptomic response of *A. aspera* occurred in three phases over a 10-day period post-injury: (i) within 2 days, (ii) at 4 days and (iii) at 10 days.

The initial response following injury was primarily TLR pathway dependent, as indicated by an upregulation of *TIR-1* 1 day post-injury, followed by increases in expression of several of its downstream signalling proteins, including *TRAF6*, *MEKK1* and *ERK2* 2 days post-injury. The limited response within the first day post-injury is consistent with an earlier report by van de Water *et al.* (2015). As AMP production is primarily regulated by the TLR signalling pathway, the observed upregulation of components of this pathway could indicate a significant production of AMPs in response to injury, which is consistent with previous short-term (septic) wounding response studies in a range of invertebrates (Fehlbaum *et al.* 1994; Rodriguez de la Vega *et al.* 2004; Pujol *et al.* 2008; Xu *et al.* 2010). Although scleractinian corals and other cnidarians are known to possess AMPs (Fraune & Bosch 2007; Fraune *et al.* 2011; Vidal-Dupiol *et al.* 2011), no AMP has been isolated from the species of *Acropora* to date. Upregulation of components in the pathway leading to AMP production suggests the likelihood that *A. aspera* produces this class of molecules.

We also found evidence of a possible positive feedback loop in the expression of TLR signalling pathway molecules in the first phase of the immune response. A positive feedback loop for TLRs has been described in mice, in which activation of TLR leads to the production of cytokines, which subsequently amplify TLR expression (Gomariz *et al.* 2005) and potentially its downstream signalling molecules. Upregulation of TLR signalling pathway genes in the early stages of wound healing and a positive feedback loop in the expression of these genes enable the coral to respond more rapidly, thereby potentially enhancing the immune response and the detection of MAMPs.

The strongest regulation of genes was observed on Day 4, and this marks the second phase, involving multiple genes of the TLR signalling pathway (*TIR-1*, *TRAF6*, *MEKK1*, *ERK2*, *cFos* and *NF-κB*) as well as immune effector genes that are part of the complement system (*Bf* and *C3*). The third phase occurred 10 days post-injury, when lesions had mostly healed, and was characterized by increases in expression levels of genes of the TLR pathway (*cFos* and *ERK2*), complement system (*Bf*) and coagulation system (*HL-2*). Together, these genes may have been involved in the maintenance of the healthy bacterial community, although a role in tissue regeneration and development of cFos and ERK2 is also probably based on their role in cell proliferation.

Expression of several coral genes displayed unexpected patterns in response to physical injury. In the case of *MAPK p38*, the downregulation indicates that it is unlikely to be involved in the wounding response in *A. aspera*. As MAPK p38 is an important mediator of the stress and immune response, this could also suggest that functional redundancy of signalling pathways may be present. The low or undetectable levels of the C-type lectin *CTL-2* in control, combined with the up to 23.5-fold significantly higher levels of this lectin in injured corals, indicate that *CTL-2* may play a role (potentially in the complement system) in the wounding-induced immune response or during the establishment of endosymbiosis in regenerating tissues. *Millectin* was constitutively downregulated for 7 days post-injury, which is in contrast with the previous reports showing that injury does not change *millectin* gene expression (Kvennefors *et al.* 2010). Ten days post-injury, *millectin* gene expression was back to normal, which could indicate that it functions in the maintenance of symbiosis with *Symbiodinium* (Kvennefors *et al.* 2008, 2010). Consistent with a previous study reporting upregulation of C3 following injury (Kvennefors *et al.* 2010), results from our study show a 1.41-fold increase in *C3* expression on Day 4. Factor B (*Bf*), part of the C3 convertase of the alternative complement pathway, which converts C3 in C3a and C3b, was also upregulated 4 days post-injury, confir-

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ming a role for this protein in the complement system. However, upregulation of *Bf* was also observed 10 days post-injury, in contrast to the downregulation of *C3*, suggesting an additional, currently unknown, role for *Bf*.

Gene expression patterns for the majority of the genes of interest were variable throughout our 10-day post-injury study period. Such variability has significant implications for the design of studies focused on detecting or characterizing invertebrate immune responses. Sampling at short intervals is required to provide sufficient resolution to gain an appropriate understanding of gene regulation in corals. A study by Schwanhausser *et al.* (2011) found that, on average, approximately 40% of protein expression levels in mammalian cells can be explained by mRNA expression, suggesting that the remaining 60% are regulated by post-transcriptional and post-translational processes. Accordingly, it is possible that immune response proteins in corals could be continuously upregulated following injury and that the observed fluctuations in differential gene expression might be due to trade-offs among energy investment options, with corals switching transcription off when protein levels are sufficient and only investing in the expression of a gene when the encoded protein levels drop below a certain threshold level.

### *The proPO system following wounding*

Upregulation of the proPO system in *A. aspera* in the first 3 days and at 10 days post-injury correlated with the first and third transcriptomic immune response phases, suggesting a positive feedback loop potentially involving components of the TLR or NLR pathway. As vesicles containing melanin and components of the proPO-activating system are rapidly released in response to wounding to seal the lesion by the formation and hardening of a clot (Palmer *et al.* 2011b), it is unlikely that the proPO system is involved in these processes at the later stage (10 days post-injury) when upregulation was observed. No increase in PO activity was observed at 4 days post-injury when the most significant transcriptomic responses occurred, further indicating that the proPO system plays other roles in addition to wound sealing in the coral immune response. A recent study found increased levels of PO activity in proliferating tissues of two scleractinian corals (D'Angelo *et al.* 2012), suggesting that the upregulation in PO activity seen in our study might be correlated with tissue regeneration and wound healing. The second lower peak in PO activity at 10 days is consistent with a role of PO in proliferating tissues, given that wound healing was not complete. Lack of upregulation of PO activity on days 4 and 7 post-injury suggests that the functioning of this system may be more complex that previously appreciated. For example, the coral may use a feedback mechanism to regulate the amount of melanin in its proliferating tissues and thus regulate the activation status of the melanization cascade. In addition, in the bumble bee *Bombus terrestris*, complex temporal PO activity patterns after injection with immunostimulants were observed until 14 days after injection (Korner & Schmid-Hempel 2004). Patterns observed in *A. aspera* could therefore also be the result of normal temporal PO activity dynamics after wounding.

### *The role of GFP-like proteins in the response to injury*

Temporal patterns in the tissue concentration of the non-fluorescent chromoprotein corresponded with those in PO activity. CP exhibits stronger antioxidant properties than FPs (Palmer *et al.* 2009), and a possible explanation for the correlation between PO activity and CP concentration is that *A. aspera* upregulates CP expression to mitigate the adverse effects of PO-produced ROS on its own tissues. However, the extent to which GFP-like proteins function as antioxidants is questionable and requires further study, as increased CP expression has been implicated to play a role in other processes such as photoprotection in accelerated tissue growth in lesions (D'Angelo *et al.* 2012). Together, these results suggest a significant role for CP in the longer-term immune response and during tissue regeneration following wounding, although its exact role remains to be elucidated.

Temporal patterns in the concentrations of fluorescent proteins suggest they may play a role in wound healing and tissue regeneration processes. The upregulation of all FP classes (CFP, GFP and RFP) observed on Day 2 post-injury, when tissues are proliferating, but lack the full *Symbiodinium* complement, corresponds with a response to increased light fluxes (D'Angelo *et al.* 2008). At later time points, however, only GFP appeared to be involved in the wounding response, but at very low levels, which is in contrast to a recent study in which CFP and RFP were implicated in coral tissue growth and wound repair in *Acropora pulchra* and *Porites lobata* (D'Angelo *et al.* 2012). The authors hypothesized that the types of GFP-like proteins in the growth zone could differ between species, which could account for our contrasting results. Upregulation of fluorescent proteins and the non-fluorescent chromoprotein at the lesion site following physical damage may occur in response to increases in the internal light field within coral tissues (D'Angelo *et al.* 2008) due to low densities of *Symbiodinium* cells and light reflection by the skeleton (D'Angelo *et al.* 2012; Smith *et al.* 2013). Thereby, this mechanism potentially protects newly formed tissue and proliferating coral cells from high light conditions (Salih *et al.* 2000; Dove *et al.* 2001), and/or *Symbiodinium* cells from

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damage to their photosystems (Smith *et al.* 2013). Strong support exists for a photoprotective function of some GFP-like proteins in corals under high environmental light intensities (Salih *et al.* 2000; Dove *et al.* 2001). Differences in the specific class of GFP-like proteins upregulated between our study and previous studies highlight the need for further research into the role of these proteins in the wounding response of corals.

### *Molecular responses involved in tissue regeneration*

Elevated expression levels of *ERK2* and *cFos* following wounding potentially link these genes with the tissue regeneration process. The regeneration process involves sealing of the lesion, followed by the development of cell layers and polyp structures through cell proliferation and differentiation. ERK2 is a protein kinase that is also activated via growth factor signalling and in turn activates the transcription factor complex AP-1, consisting of cFos and cJun, as well as ATF family members (Hess *et al.* 2004). In contrast to the upregulation of *ERK2* and *cFos*, *cJun* expression remained stable, whereas *ATF4/5* was downregulated. As cFos and cJun form the AP-1 complex, it is unlikely that cJun is not involved in cell proliferation and differentiation during wound healing. Upregulation of cJun protein levels might still have been possible, for example as a result of an increased stability of mRNA transcripts allowing increased protein expression, in the absence of increased gene expression (Wu & Brewer 2012). In addition, cJun and cFos only form the AP-1 complex while being in a phosphorylated state, and a consistent expression level but increased phosphorylation status could result in an increased AP-1 activity during wound healing (Hess *et al.* 2004). Therefore, a role for cJun in the regeneration process cannot be excluded and further proteomic approaches are required to address these expression patterns.

Coral tissue regeneration requires not only the differentiation and development of host tissues, but also the establishment of symbiosis with endosymbiotic *Symbiodinium* algae in the newly formed gastrodermis through *Symbiodinium* recolonization from surrounding tissues. Previous studies suggest that the immune effectors of the lectin–complement pathway, C3 and millectin, are implicated in the maintenance of *Symbiodinium* endosymbiosis in *Acropora millepora*, in addition to playing a role in the antimicrobial response (Kvennefors *et al.* 2008, 2010). Upregulation of components of the lectin–complement pathway, such as *C3* and *Bf* as well as the lectin *CTL-2*, observed in this study, indicates a role of this pathway in the tissue regeneration process that may be related to the establishment of *Symbiodinium* symbiosis in newly differentiated host cells. However, the *millectin* homologue was consistently downregulated in our study until 7 days post-injury. In combination with the results of these previous studies, our results suggest that, whereas other components of the lectin–complement pathway may be involved in the establishment of symbiosis, millectin may be involved in the maintenance of the symbiosis. This interpretation is supported by the increases in expression to control levels observed 10 days post-injury, when lesions were almost completely healed and *Symbiodinium* density-based tissue pigmentation returned to normal.

### *Bacterial community responses following physical damage*

We did not observe any changes in bacterial abundance or community composition in healthy or injured corals throughout our study period. Bacteria play a major role in the health of the coral holobiont, and it is therefore highly advantageous for the coral to maintain a healthy beneficial microbial community. Several invertebrates, including cnidarians, have been shown to control the composition of their microbial community through the expression of antimicrobial peptides (AMP) (Fraune & Bosch 2007; Kounatidis & Ligoxygakis 2012). The recent isolation of an AMP from *Pocillopora damicornis* suggests this may also hold true for corals. Furthermore, Toll-like receptors fulfil an essential role in the detection and regulation of the bacterial community composition in the cnidarian *Hydra* (Franzenburg *et al.* 2012) and are the main regulators of AMP expression. As the putative Toll-like receptor TIR-1 and its downstream signalling molecules were upregulated in *A. aspera* following injury, we speculate that this coral species is capable of maintaining a healthy bacterial community throughout the repair process of an artificially inflicted wound via TLR-regulated AMP expression.

In this study, we demonstrated that the coral *A. aspera* exhibits a strong immune response following physical injury, likely preventing bacterial infection and enabling wound healing within 10 days. Overall, the physical damage response consisted of three phases, which are summarized in Fig. 6. During the first phase, the coral activates the immediate response mechanisms, such as the melanization cascade to seal the lesion and prevent microbial infestation of the tissues (Fig. 6, *Phase I*). Increased expression of components of the TLR pathway suggests potential regulation of the production of AMPs via a positive feedback loop. Elevated AMP production may allow the coral to maintain a healthy bacterial community that simultaneously produces antimicrobial compounds, thereby preventing pathogenic microbes from establishing. Simultaneously, wound repair and tissue regeneration processes commence, including increased cell proliferation (*ERK2* and its downstream transcription

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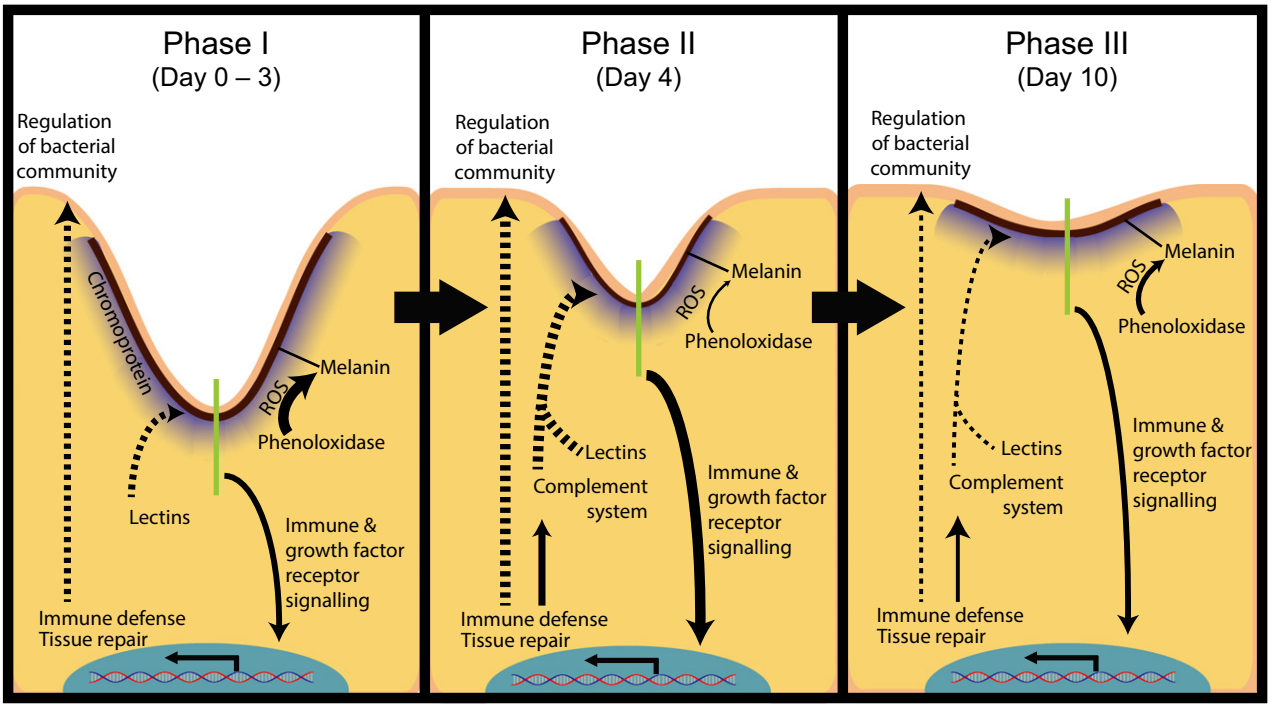


**Fig. 6** Overview of the different phases in coral immune response following physical damage. *Phase I* is characterized by rapid upregulation of the photoprotective chromoprotein and increased phenoloxidase activity that forms a melanin barrier preventing microbes from entering the lesion. Immune receptor signalling is increased and potentially regulates the coral-associated bacterial community via antimicrobial peptides (AMP). In addition, lectins eliminate invading microbes. Growth factor receptor signalling initiates tissue repair processes resulting in cell proliferation and differentiation. *Phase II* shows a marked increase in the expression of immune and growth factor receptor pathway signalling components that may result in increased AMP production and rapid tissue repair. Besides, the complement system and lectins are involved in the response to prevent microbes from infecting tissues. During *Phase III*, the lesion has nearly recovered and tissue repair and remodelling processes continue while the immune system maintains the bacterial community. Some immune system components may facilitate the recolonization of the tissues by *Symbiodinium*. Thickness of arrows indicates the strength of the response. Dashed lines indicate potential defence processes regulated by the immune system.

factors). However, at the lesion front, these highly proliferating cells with their endosymbionts are exposed to high (UV) light radiation intensity. As a result, corals increase the expression of photoprotective GFP-like proteins in these tissues, likely to protect the highly proliferating cells and *Symbiodinium* from high light intensity. The most significant response occurs 4 days post-injury, marking the second phase (Fig. 6, *Phase II*). Strong signalling via the TLR pathway stimulates the immune system further, resulting in the activation of additional immune effector mechanisms such as microbe-agglutinating lectins and the complement pathway, as well as regulation of the bacterial community. Tissue repair processes also remain activated until lesions have completely recovered during the third phase of the wounding response (Fig. 6, *Phase III*). At this time, the coral immune system still successfully maintains a healthy bacterial community via TLR pathway-induced increased AMP expression. In addition, symbiosis with *Symbiodinium* in newly formed tissues is established and maintained using the lectin–complement pathway completing the recovery.

Overall, the response following injury and during wound repair in *Acropora aspera* is dynamic, consisting of multiple phases, and is characterized by increased expression of multiple immune system-related genes, phenoloxidase, as well as GFP-like proteins over time. Together, these processes enable the coral to maintain a healthy bacterial community, prevent microbial infection and thereby support rapid recovery from physical damage.

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J.W., W.L. and T.D.A. designed and conducted the experiment. J.W. analysed the samples and the data. J.W., T.D.A., W.L., D.G.B., B.L.W. and M.O. wrote the manuscript.

## Data accessibility

Immune parameter expression and activity data: Dryad doi: 10.5061/dryad.r24b1.

Bacterial 16S rRNA amplicon data and analysis files: Dryad doi: 10.5061/dryad.r24b1.

Bacterial 16S rRNA amplicon sequences: NCBI SRA Accession no. SRP052969.

## Supporting information

Additional supporting information may be found in the online version of this article.

**Fig. S1** *Acropora aspera* colony.

**Fig. S2** (a) Temporal patterns in total potential phenoloxidase activity (tpPO) in response to injury.

**Table S1** Sample overview.

**Table S2** Overview immune genes and GeXP primers.

**Table S3** Overview of statistical analysis of main effects.

**Table S4** Overview of statistical analysis outcome of Least Significant Difference post-hoc tests.

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